Evaluating AI Agent Teams with Decentralized Information

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Motivation and Goal

An AI agent team may share a decision objective while its members have access to different records. In a financial review, one agent may receive a cash balance, another a restriction on its use, and a third a repayment deadline. Reaching a correct decision requires members to find, report, and combine these facts. Individual model performance does not establish whether these steps succeed.

Team decision theory and decentralized control study such cooperation under information constraints [1,2]. Decentralized information refers to agents making decisions from information that differs across members. The information structure specifies which observations, past actions, and messages each agent can use at each step. Communication changes the information available to later decision makers. We will hold the decision problem fixed while changing information access and communication. For Arena, which evaluates model choices and task cost [3,4], this will connect model comparisons to the evidence agents retrieve, exchange, and correctly account for in a decision.

How can we evaluate and improve AI agent teams when the information needed for a decision is decentralized?

Thrust 1 Evaluating Decisions under Decentralized Information

Construct decisions with checked evidence. Building on our ongoing prototype, we will construct tasks from financial documents and software repositories. For each task, we will identify pivotal facts that together support a reference decision. Removing any selected fact must leave that decision unsupported under our checks. We will examine alternative evidence and independently review whether an agent's local records, together with shared information, already support the full answer. These checks make changes in information access meaningful rather than merely changing document length.

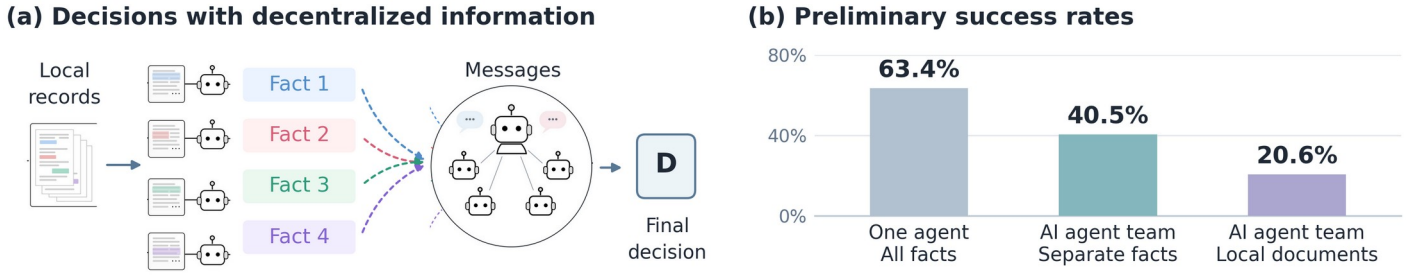


Figure 1. (a) Agents extract and exchange evidence before a final decision. (b) Preliminary success rates across eight models and twelve source corpora. Rates average models within each corpus, then weight corpora equally. The two AI agent team conditions use the same fact allocation, communication rules, and resource limits.

Separate finding information from communicating it. We will compare one agent receiving all pivotal facts, an AI agent team receiving those facts separately, and the same team receiving source documents containing its assigned facts. The communication gap is the drop in success when the facts are split across agents. The retrieval gap is the further drop when they must extract those facts from documents (Figure 1). A fourth baseline will give one agent all source records, retaining document retrieval while removing inter-agent communication. On selected tasks, we will vary whether a fact goes to one agent or all agents, whether most facts go to one member, and whether an additional member holds no pivotal fact.

Evaluate communication rules and model choices. Members will speak in a fixed order, report without seeing one another's messages, or respond to a coordinator that selects whom to query and makes the final decision. On matched task subsets, we will compare two communication rules under two allocations of the same facts and measure how the rules' success-rate difference changes between allocations. We will compare six models, including replacing a participant or coordinator while keeping its partners fixed. Comparisons will match total limits on tokens read and generated, and report actual team cost, completion, and accuracy. Resource-limit failures count as unsuccessful outcomes, with service outages reported separately.

Completion means returning a valid decision within the resource limit. Uncertainty estimates will resample whole source documents or repositories, preserving comparisons on identical tasks.

Thrust 2 Diagnosing Failures and Validating Improvements

Diagnose where relevant information is lost. We will check which pivotal facts appear in each agent's extracted evidence and messages, and whether the final decision correctly accounts for them. We will freeze a conversation and use the same final-decision model after adding a correct inference supported by its existing facts, without supplying the answer, or removing a statement that relies on an earlier error while retaining the original error. We will also hide messages from members assigned no pivotal facts, retaining all fact-holders' reports, to test whether such messages help or obstruct the final decision. Separately, we will fix participant selection, speaking order, sources, and budgets while comparing original coordinator queries, queries augmented with topic labels and an instruction to use them, and a broad invitation to contribute. Labels name evidence categories without revealing private facts or reference answers. This tests changes in questions and added topic guidance with participant selection fixed.

Use training to validate the evaluation. Training lets us test whether a higher final score reflects better collaboration. Building on initial trials, we will use reinforcement learning with a reward for the final decision. With a coordinator, we will compare training participants, the coordinator, and both together. Without one, we will train a shared participant model to choose messages, the next speaker, and when to decide. We will test whether gains in the final score coincide with better information coverage and use, without separately rewarding a prescribed communication pattern. We will separate returning a valid decision from answering correctly, so higher completion alone is not counted as improved evidence integration.

Test whether improvements carry over. We will conduct three independent training runs and evaluate on new source documents, team sizes, and allocations of facts. For tasks completed by both original and trained teams, we will have both final-decision models answer from each fixed conversation, preserving the final speaker's identity, assigned information, and prompt. This two-by-two comparison tests whether gains follow the conversation, the final-decision model, or their combination. Separately, a fixed model will read all conversations, including valid messages from unfinished runs, to avoid hiding completion failures. Preliminary results show that wider participation need not transmit more pivotal facts on a new corpus. We will report participation, fact coverage, completion, and accuracy. On selected tasks, we will vary message limits across 32, 64, 128, and 256 tokens and record how often limits bind.

Evaluation Plan and Outputs

During months 1–3, we will independently audit approximately 200 finance and software tasks and establish their evidence and scoring rules. Months 4–6 will compare the six models under controlled information access and communication rules. Months 7–9 will test communication changes and the focused training study. Months 10–12 will evaluate transfer and release the results. Protocol and budget variations will use selected task subsets. Training will use one pretrained base model, with separate copies for participant and coordinator roles. Related tasks from the same document or repository will remain together when separating training and evaluation data. Final decisions will be checked against reference answers, and code changes will use repository tests. Human review will check a sample of automated scores. We will release permitted tasks, evaluation code, and annotated conversations, with accuracy, uncertainty, and total team cost. Arena collaboration can guide task coverage and reporting.

We request $50,000 over 12 months: $30,000 for partial graduate research support and benefits, $12,000 for model runs and computing, $5,000 for independent checks, and $3,000 for releasing code and sharing results. Final allocations will follow institutional gift policy.

Selected References

[1] Ho and Chu. Team Decision Theory and Information Structures in Optimal Control Problems—Part I. IEEE TAC, 17(1):15–22, 1972.

[2] Nayyar, Mahajan, and Teneketzis. Decentralized Stochastic Control with Partial History Sharing: A Common Information Approach. IEEE TAC, 58(7):1644–1658, 2013.

[3] Arena. Agent Arena: Causal Evaluation of Agents in the Real World, 2026. arena.ai/blog/agent-arena-methodology.

[4] Arena. Agent Leaderboard Improvements: Categories & Task Cost, 2026. arena.ai/blog/agent-categories-and-cost.